



The remaining time

A literal countdown clock to disaster, covering both the 1.5 degree and 2 degree Celsius scenarios. <http://digit.in/aug21-56>



Estimating the budget

The report in Nature that used sophisticated modelling to estimate the budget. <http://digit.in/aug21-57>

FINDING EARTH'S CARBON BUDGET



The clock is ticking.
But how fast?

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Earth's delicate ecosystem is in danger, affecting all life down to the critters literally

living beneath a rock. The problem is the buildup of carbon dioxide and other greenhouse gases in the atmosphere, increasing to a global rise in temperatures because of industrial activity.

The question of figuring out exactly how long we have before there is a catastrophic impact on the planet is a difficult one to answer, but one that can be done with numbers!

The Earth's carbon budget is a mathematical estimation by which every involved in working on conserving the planet's ecosystem gets a deadline before everything comes crashing down. While it might not be an all inclusive formula that gives gives us a specific date for the impending doomsday, but sufficient to give everyone a window to aim for. This is an important calculation to drive policy decisions, which

are urgently required to reduce carbon dioxide emissions, the biggest culprit in Earth's climate change related issues.

WHAT IS IT EXACTLY?

As defined by the World Meteorological Organization, 'A global carbon budget determines the input of carbon dioxide to the atmosphere by emissions from human activities, balanced by output (storage) in the carbon reservoirs on land or in the ocean.' In simpler words, it is an estimate of the threshold of Carbon dioxide that we can afford to have in our atmosphere before severe climatic changes and alterations start to kick in, with the planet becoming inhabitable, devastating all life on Earth.

The quest for figuring out this budget began when the leaders of the world met and signed the historic Paris Agreement. The Paris Agreement is a "legally binding international treaty on climate change". Signed by a total of 196 parties, this agreement sets a goal for each signing party to strive towards stopping global warming "to well below 2, preferably to 1.5 degrees Celsius, compared to pre-industrial levels". This means that each party that has signed this agreement will have to take specific practical steps to ensure that the

temperature of the Earth stays below the recommended levels.

There are several clauses to this agreement that make sure that in whatever capacity they can, countries make efforts to stop the emissions of carbon dioxide. Today, the biggest emitters of carbon dioxide in the world are located in Asia, followed by North America, Europe, and the rest of the world. After looking at these numbers, in order to determine how close are to the goalposts, and how much amount of carbon dioxide emission needs to be cut down, we use something known as the carbon budget, which is calculated every year, tracking the changes in the required metrics for carbon dioxide emissions on Earth.

HOW DO WE CALCULATE IT?

There is no single organization that publishes the annual numbers for giving you the Earth's carbon budget. Several organizations all over the world get their hands on whatever data that they can, which is in turn collated to form the annual report stating how much time there is left for us to take steps before the planetary ecosystem collapses.

What started out as a simple tool to draft out climatic models has now morphed into being a much